COMPUTER SCIENCE + COMPUTER ENGINEERING FOR GAMES

09

VOLUME NINE

PACKAGE 10 / GRADUATE SPECIALIZATION

Primary Reading Guide

Preserved program references, focused graduate sources and terminology

55 INHERITED REFERENCES / 17 FOCUSED ADDITIONS

Edition: September 28, 2026

Original lesson sequence preserved. Self-study materials and runnable educational references; not an accredited degree or completed learner submission.

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Primary reading directory

R01-R55 retain the original program IDs. S01-S17 are focused additions for this expansion. Citations such as [S05] refer to the entries below; equations and worked cases not attributed to another source are original teaching derivations. External resources retain their own rights and are linked rather than redistributed. Moving documentation is not a promise of compatibility with an untested installed version. Record exact versions for your own artifact.

R01 — CS2023 knowledge areas

ACM / IEEE-CS / AAAI

https://csed.acm.org/knowledge-areas/

Breadth checklist, not a claim of accreditation.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R02 — Computer Engineering Curricula 2016

ACM / IEEE Computer Society

https://ieeecs-media.computer.org/assets/pdf/ce2016-final-report.pdf

Computer-engineering breadth and laboratory expectations; historical curriculum report.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R03 — CS50 Introduction to Programming with Python

Harvard University

https://cs50.harvard.edu/python/

Beginner lectures and practice; complete exercises rather than only watching.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R04 — Python tutorial and language documentation

Python Software Foundation

https://docs.python.org/3/tutorial/

Reference companion; the tutorial itself assumes some programming familiarity.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R05 — Pro Git

Scott Chacon and Ben Straub

https://git-scm.com/book/en/v2

Version control and collaboration.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R06 — 6.042J Mathematics for Computer Science

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-042j-mathematics-for-computer-science-spring-2015/

Proofs, graphs, counting and discrete probability.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R07 — 18.06 Linear Algebra

MIT OpenCourseWare

https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/

Matrices, vector spaces, eigenvalues and problem sets.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R08 — 18.01SC Single Variable Calculus

MIT OpenCourseWare

https://ocw.mit.edu/courses/18-01sc-single-variable-calculus-fall-2010/

Calculus lectures, exercises and assessments.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R09 — 18.05 Introduction to Probability and Statistics

MIT OpenCourseWare

https://ocw.mit.edu/courses/18-05-introduction-to-probability-and-statistics-spring-2022/

Probability, inference and quantitative reasoning.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R10 — 6.003 Signals and Systems

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/

Sampling, convolution, transforms and systems.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R11 — C++ Core Guidelines

C++ Core Guidelines editors

https://isocpp.github.io/CppCoreGuidelines/CppCoreGuidelines

Resource safety, interfaces, classes and concurrency; guidance rather than the normative language standard.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R12 — CMake tutorial

Kitware

https://cmake.org/cmake/help/latest/guide/tutorial/index.html

Build systems; use documentation matching the installed release.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R13 — 6.006 Introduction to Algorithms

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/

Data structures, graph algorithms and dynamic programming.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R14 — Kaleidoscope and LLVM tutorials

LLVM Project

https://llvm.org/docs/tutorial/

Compiler frontends, intermediate representation and JIT compilation.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R15 — 6.004 Computation Structures

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-004-computation-structures-spring-2017/

Logic, processor design, memory and systems.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R16 — Operating Systems: Three Easy Pieces

Remzi and Andrea Arpaci-Dusseau

https://pages.cs.wisc.edu/~remzi/OSTEP/

Virtualization, concurrency and persistence; book by its authors.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R17 — RFC 9293: Transmission Control Protocol

IETF / RFC Editor

https://www.rfc-editor.org/rfc/rfc9293.html

Transport semantics; reference reading after networking foundations.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R18 — PostgreSQL tutorial

PostgreSQL Global Development Group

https://www.postgresql.org/docs/current/tutorial.html

Relational data, transactions and SQL; pin local server and client versions.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R19 — 15-462/662 Computer Graphics

Carnegie Mellon University

https://15462.courses.cs.cmu.edu/fall2021/

Rendering, geometry and animation course structure and exercises.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R20 — 15-466/666 Computer Game Programming

Carnegie Mellon University

https://graphics.cs.cmu.edu/courses/15-466/

Runtime systems and game-production programming.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R21 — Vulkan tutorial

Khronos Group

https://docs.vulkan.org/tutorial/latest/00_Introduction.html

Explicit graphics API study; feature-check the device before using the moving latest tutorial.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R22 — SDL3 documentation

SDL project

https://wiki.libsdl.org/SDL3/FrontPage

Windowing, input, audio and platform services; do not mix SDL2 and SDL3 examples.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R23 — Physically Based Rendering, fourth edition

Matt Pharr, Wenzel Jakob, Greg Humphreys

https://www.pbr-book.org/4ed/contents

Rendering mathematics and implementation; use chapter titles in lesson assignments.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R24 — Direct3D 12 graphics

Microsoft

https://learn.microsoft.com/en-us/windows/win32/direct3d12/direct3d-12-graphics

Windows graphics backend reference.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R25 — Metal developer resources

Apple

https://developer.apple.com/metal/

Apple-platform graphics and compute reference.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R26 — Programming with C++ in Unreal Engine

Epic Games

https://dev.epicgames.com/documentation/en-us/unreal-engine/programming-with-cplusplus-in-unreal-engine

Engine comparison after C++ foundations, not a beginner substitute.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R27 — Fix Your Timestep!

Glenn Fiedler

https://gafferongames.com/post/fix_your_timestep/

Original practitioner explanation of fixed-step simulation; read critically and implement your own lab.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R28 — Box2D documentation

Box2D project

https://box2d.org/documentation/

Collision, constraints and simulation reference.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R29 — Xbox Accessibility Guidelines

Microsoft

https://learn.microsoft.com/en-us/xbox/accessibility/guidelines

Accessibility design and testing; not a legal-compliance certificate.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R30 — 6.002 Circuits and Electronics

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-002-circuits-and-electronics-spring-2007/

Circuits, transistors and laboratory reasoning.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R31 — Nand to Tetris

Noam Nisan and Shimon Schocken

https://www.nand2tetris.org/

Computer construction projects; review the authors' license terms.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R32 — Verilator user guide

Verilator project

https://verilator.org/guide/latest/

RTL simulation and linting; match supported SystemVerilog subset.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R33 — Yosys documentation

YosysHQ

https://yosyshq.readthedocs.io/projects/yosys/en/latest/

Synthesis and technology mapping; stable installed release preferred over a development snapshot.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R34 — cocotb documentation

cocotb project

https://docs.cocotb.org/en/stable/

Python-based HDL verification; simulator and version compatibility must be checked.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R35 — RISC-V ISA specifications

RISC-V International

https://docs.riscv.org/reference/isa/

Instruction-set reference; record the exact ISA version and implemented extensions.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R36 — KiCad documentation

KiCad project

https://docs.kicad.org/

Schematic capture, PCB layout and manufacturing outputs.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R37 — CUDA Programming Guide

NVIDIA

https://docs.nvidia.com/cuda/cuda-programming-guide/

GPU execution and memory; CUDA is an optional vendor-specific lab route.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R38 — gem5 documentation

gem5 project

https://www.gem5.org/documentation/

Architecture simulation; distinguish model outputs from measured hardware results.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R39 — OpenROAD documentation

OpenROAD project

https://openroad.readthedocs.io/en/latest/

RTL-to-physical-design study; educational results are not production signoff.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R40 — Precalculus, second edition

OpenStax / Rice University

https://openstax.org/details/books/precalculus-2e

Algebra, functions and trigonometry bridge before calculus.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R41 — 18.02SC Multivariable Calculus

MIT OpenCourseWare

https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/

Partial derivatives, gradients, multiple integrals and constrained differentials.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R42 — Convex Optimization

Stephen Boyd and Lieven Vandenberghe

https://web.stanford.edu/~boyd/cvxbook/

Graduate optimization text available from its authors; copyrighted, linked rather than redistributed.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R43 — Roofline performance model, technical report

Samuel Williams, Andrew Waterman and David Patterson

https://www2.eecs.berkeley.edu/Pubs/TechRpts/2008/EECS-2008-134.html

Original 2008 performance-model reference; use current measured hardware limits for experiments.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R44 — Spatiotemporal reservoir resampling for real-time ray tracing with dynamic direct lighting

Benedikt Bitterli and coauthors

https://research.nvidia.com/publication/2020-07_spatiotemporal-reservoir-resampling-real-time-ray-tracing-dynamic-direct

Original ReSTIR research, 2020; example reproduction target, not a claim to be the latest method.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R45 — AddressSanitizer documentation

LLVM Project

https://clang.llvm.org/docs/AddressSanitizer.html

Memory-error instrumentation, limitations and supported toolchains.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R46 — CS188 Introduction to Artificial Intelligence textbook

University of California, Berkeley

https://inst.eecs.berkeley.edu/~cs188/textbook/

Search, games, decision processes, learning and probabilistic reasoning.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R47 — Dive into Deep Learning

Aston Zhang, Zachary Lipton, Mu Li and Alexander Smola

https://d2l.ai/

Learning fundamentals, gradients, generalization and reproducible exercises; pin compatible framework versions.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R48 — 6.823 Computer System Architecture

MIT OpenCourseWare

https://ocw.mit.edu/courses/6-823-computer-system-architecture-fall-2005/

Graduate architecture foundations; historical course concepts supplemented with current specifications.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R49 — ECE5745 Complex Digital ASIC Design

Christopher Batten / Cornell University

https://www.csl.cornell.edu/courses/ece5745/

Graduate ASIC methodology, circuits, EDA and design-project expectations.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R50 — SymbiYosys documentation

YosysHQ

https://yosyshq.readthedocs.io/projects/sby/en/latest/

Formal hardware verification, safety, liveness and cover properties.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R51 — Driver implementer API guide

Linux kernel contributors

https://docs.kernel.org/driver-api/index.html

Device ownership, I/O, DMA and driver interfaces; use disposable VMs or simulated devices for kernel experiments.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R52 — OpenXR registry and specifications

Khronos Group

https://registry.khronos.org/OpenXR/

XR coordinate spaces, frame timing and runtime interfaces; optional hardware-dependent extension.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R53 — Zephyr documentation

Zephyr Project

https://docs.zephyrproject.org/latest/introduction/index.html

Embedded kernel, drivers, board support and simulation; select and pin a supported release.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R54 — 18.085 Computational Science and Engineering I

MIT OpenCourseWare

https://ocw.mit.edu/courses/18-085-computational-science-and-engineering-i-fall-2008/

Graduate numerical methods, equilibrium systems, differential equations and variational reasoning.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

R55 — Author publication archive: position-based simulation and XPBD

Matthias Mueller and collaborators

https://matthias-research.github.io/pages/publications/publications.html

Read the named 2016 XPBD paper and related course notes for the simulation reproduction; papers retain their own copyrights.

Source status: Inherited from the original program; not every moving page was revalidated for this expansion.

S01 — An Introduction to the Conjugate Gradient Method Without the Agonizing Pain (1994)

Jonathan Richard Shewchuk

https://www.cs.cmu.edu/~quake-papers/painless-conjugate-gradient.pdf

PCG derivation and Appendix B3; PDF and algorithm page inspected.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S02 — Convex Optimization (2004), author-hosted text

Stephen Boyd and Lieven Vandenberghe

https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf

Convexity, optimality and numerical methods; KKT page inspected. Also R42.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S03 — Parallel Prefix Sum (Scan) with CUDA, GPU Gems 3, Chapter 39

Mark Harris, Shubhabrata Sengupta and John D. Owens

https://developer.nvidia.com/gpugems/gpugems3/part-vi-gpu-computing/chapter-39-parallel-prefix-sum-scan-cuda

Work-efficient scan foundations; historical algorithms, not current tuning defaults.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S04 — CUDA Programming Guide

NVIDIA

https://docs.nvidia.com/cuda/cuda-programming-guide/

Execution and memory contracts; moving documentation consulted, no matching toolkit/device validation claimed.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S05 — XPBD: Position-Based Simulation of Compliant Constrained Dynamics (2016)

Miles Macklin, Matthias Mueller and Nuttapong Chentanez

https://matthias-research.github.io/pages/publications/XPBD.pdf

Original paper; multiplier/compliance equations inspected. A bounded reproduction target, not a completed result supplied here.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S06 — PBRT 4e: Mapping Path Tracing to the GPU

Matt Pharr, Wenzel Jakob and Greg Humphreys

https://www.pbr-book.org/4ed/Wavefront_Rendering_on_GPUs/Mapping_Path_Tracing_to_the_GPU

Wavefront organization and queue-oriented rendering.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S07 — PBRT 4e: Improving Efficiency

Matt Pharr, Wenzel Jakob and Greg Humphreys

https://www.pbr-book.org/4ed/Monte_Carlo_Integration/Improving_Efficiency

Importance sampling, MIS and roulette.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S08 — PBRT 4e: A Better Path Tracer

Matt Pharr, Wenzel Jakob and Greg Humphreys

https://www.pbr-book.org/4ed/Light_Transport_I_Surface_Reflection/A_Better_Path_Tracer

Light sampling and emitter-hit weighting; compare against the bounded educational implementation.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S09 — Vulkan specification: Acceleration Structures

Khronos Group

https://docs.vulkan.org/spec/latest/chapters/accelstructures.html

Capability, update and resource contracts. Moving specification, not a tested API backend.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S10 — Gradient-based optimization tutorial

Mitsuba project

https://mitsuba.readthedocs.io/en/stable/src/inverse_rendering/gradient_based_opt.html

Primary implementation documentation for inverse rendering; no Mitsuba installation required by the portable references.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S11 — In Search of an Understandable Consensus Algorithm, extended version

Diego Ongaro and John Ousterhout

https://raft.github.io/raft.pdf

Replicated state-machine assumptions; PDF and architecture page inspected. The teaching sink does not implement Raft.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S12 — Safety, Liveness, and Fairness (2019)

Leslie Lamport

https://lamport.azurewebsites.net/tla/safety-liveness.pdf

Precise definitions and fairness assumptions; safety/liveness page inspected.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S13 — LLVM Language Reference Manual

LLVM Project

https://llvm.org/docs/LangRef.html

SSA, phi and arithmetic semantics. Record the LLVM version actually used in a learner experiment.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S14 — Artifact Review and Badging, version 1.1

ACM

https://www.acm.org/publications/policies/artifact-review-and-badging-current

Official search index and ACM terminology page verified; direct policy-page fetch returned 403. Supplement: https://www.acm.org/publications/badging-terms . This curriculum awards no ACM badge.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S15 — Auto-Vectorization in LLVM

LLVM Project

https://llvm.org/docs/Vectorizers.html

Loop/SLP vectorization, legality and diagnostics; primary compiler documentation.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S16 — LLVM Tutorial: Table of Contents

LLVM Project

https://llvm.org/docs/tutorial/

Frontend, IR and JIT tutorials. The bundled DSL stops at interpretation and source emission.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

S17 — Consul session semantics

HashiCorp

https://developer.hashicorp.com/consul/docs/automate/session

Session/locking behavior and authority limitations; primary service documentation, not a service deployment requirement.

Source status: Focused primary source consulted September 28, 2026; see note for retrieval limits.

Evidence and terminology quick reference

The definitions below are concise course reminders. Consult the assigned primary sources for formal assumptions and complete derivations.

Ablation

A controlled removal or replacement intended to isolate a component's contribution.

Arithmetic intensity

Useful operations per byte moved at a specified memory level; the level and counting convention matter.

Condition number

A measure of sensitivity of a mathematical problem; for invertible A in the 2-norm, ||A||2*||A^-1||2.

Critical path

The longest dependent chain in the task graph, including required waits and movement.

Disocclusion

A surface becomes newly visible, making some prior pixel history invalid.

Evidence tier

An explicit label distinguishing derivation, analytical model, software experiment, device measurement and independent review.

Fencing token

An advancing authority version checked at the receiver that commits an effect, under a specified coordination protocol.

Forward error

Distance between a computed solution and the true/reference solution, distinct from equation residual.

Identifiability

Whether the observations distinguish the unknown parameters under the chosen model.

Independent trial unit

The run, scene, device or other unit treated as independent for a specified inference.

KKT conditions

Primal/dual feasibility, stationarity and complementary slackness; necessity and sufficiency depend on assumptions.

Legality

Preservation of the accepted language's observable behavior under a stated transformation contract.

Liveness

An eventual-progress property under the relevant environment assumptions.

Multiple importance sampling

Combination of sampling strategies with weights/densities that match the estimator and common measure.

Phi node

SSA merge representation that selects an incoming value according to the predecessor edge.

Preconditioner

An auxiliary operation intended to improve iterative convergence while preserving the target system when correctly applied.

Residual

Equation mismatch, such as b-A*x, with an explicitly stated norm.

Safety

Exclusion of forbidden behavior; a finite bad prefix can demonstrate a violation.

Span

Longest dependency chain of an algorithm under a defined computation model.

Specialization

Use of known parameters to simplify a computation, trading runtime work for preparation/variant costs.

SSA

Static single assignment: each represented value has one defining assignment, with merge semantics made explicit.

Truncation bias

A change in the target expectation caused by an uncompensated finite cutoff or approximation.

Work

Total operation count under the selected algorithmic model, not elapsed time.

XPBD

A compliant constraint formulation with timestep-scaled compliance and accumulated multipliers; its assumptions and discretization remain important.